

Rowan Leonard

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EDUCATION

University of Waterloo, *BASc in Electrical Engineering*

Apr 2029

TECHNICAL SKILLS

Languages & HDLs: C/C++, Python, Verilog, VHDL, RISC-V Assembly

Tools: XSchem, ngspice (SKY130 PDK), Cocotb, Git, Linux, AutoCAD, AGI32

PERSONAL PROJECTS

SNN Accelerator, *Verilog, Cocotb*

- Designed a 16-bit spiking neural network (SNN) hardware accelerator targeting a Xilinx Zynq-7020 SoC, implementing Q1.14 fixed-point arithmetic mapped to DSP48E1 slices for area-efficient multiply-accumulate operations
- Architected four configurable engine modes – weighted accumulation, threshold/fire/reset, STDP weight updates, and exponential decay – with BRAM-based storage for synaptic weights and neuron traces
- Verified the design with a full Cocotb-based Python testbench suite, resolving edge cases in packed-array indexing, bit ordering, and clock-domain control
- Currently engineering an analog–digital hybrid machine-learning computer focused on accelerating neural computation and reducing power consumption through novel circuit-level architectural strategies

Machine Learning Library, *C++, RNNs*

- Developed a complete neural network library in C++ using only the STL, supporting research into multi-network integration, weighted residual learning, Hopfield networks, radial basis function networks, and map-based classification architectures
- Designed a network-generation algorithm that discovers high-performance architectures by optimizing training speed, parameter efficiency, and error minimization, and built supporting tooling for model serialization, dataset generation, and discrete-weight support

Computer Design, *Logisim, Custom ISA*

- Designed complete 8-bit and 16-bit computer architectures from the ground up using Logisim, modeling all schematics with custom logic-gate layouts, buses, and control circuitry for functional computation
- Developed a custom instruction set architecture (ISA), assembly language, ALU design, and control-flow system, enabling fully programmable computation on self-designed hardware with architectural flexibility

EXPERIENCE

RR Power Consulting Inc., *Power Systems Electrical Technologist*

Jan – Apr, Sep – Dec 2025

- Drafted distribution infrastructure using AutoCAD, creating and importing survey base maps
- Modeled and designed street lighting to meet individual client design criteria using AGI32 Lighting Software
- Performed voltage drop calculations and developed Excel-based tools to automate pole-height lookup and optimize street lighting design

EXTRACURRICULAR ACTIVITIES

University of Waterloo ASIC Design Club, *Member – Analog & Digital Teams*

- Designed a two-stage Miller-compensated operational transconductance amplifier (OTA) in the SKY130 PDK using XSchem and ngspice, meeting specifications of ≥ 20 dB gain, ≤ 5 mV input offset, ≥ 40 dB CMRR, and < 5 mW power draw at 1.8 V
- Contributed to digital RTL design projects as a member of the club's digital design team